

CHINONSO NGWU

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with strong specialization in Edge AI and Computer Vision systems. Build production-scale camera intelligence features end-to-end across C++ firmware on Embedded Linux (Yocto), backend (Ruby on Rails), and frontend (React), deployed across hundreds of thousands of Cisco Meraki devices. Strong expertise in computer vision, on-device ML, and shipping enterprise features from prototype to General Availability. M.Sc. from École Polytechnique.

TECHNICAL SKILLS

Languages: C, C++, Ruby, Python, JavaScript/TypeScript, Java, SQL, Bash

Frameworks: Ruby on Rails, React, Redux, Flask, Node.js, PyTorch, OpenCV, REST APIs

Embedded & Systems: Embedded Linux, Yocto, Buildroot, Linux Kernel, Cross-Compilation, Memory Management

Computer Vision & ML: YOLO, DeepSORT, ByteTrack, Multi-Object Tracking, Model Quantization, License Plate Recognition, Edge AI Deployment

Cloud & DevOps: Docker, Kubernetes, Jenkins, GitLab CI/CD, AWS, Git, Gerrit

Databases: PostgreSQL, MySQL, Redis

Practices: Agile/Scrum, Design Reviews, Code Reviews, TDD, Cross-Functional Collaboration

Spoken Languages: English (Native), Igbo (Native), French (A2)

PROFESSIONAL EXPERIENCE

Cisco Meraki | Software Engineer, MV Enterprise Experience Team

Paris, France | January 2025 – Present

- Led architecture and delivery of an embedded License Plate Recognition capability from prototype through General Availability, owning C++ on-device implementation tuned for low-power inference on Meraki camera SoCs, the Rails and React product surfaces, and post-launch reliability.
- Drive enterprise camera features end-to-end across firmware (C/C++, Embedded Linux), backend (Ruby on Rails), and dashboard (React), shipping to enterprise customers worldwide.
- Own firmware architecture improvements and lead new hardware platform bring-up on the MV camera line.
- Mentor engineers and lead design reviews; partner with QA, Product, and Architecture to scope and deliver features against enterprise SLAs.

Cisco Meraki | Software Engineer, Camera Intelligence Team

Paris, France | April 2023 – December 2024

- Developed firmware features for object detection and multi-object tracking in C++ on resource-constrained edge hardware, deployed to hundreds of thousands of cameras in production.
- Built an internal ML Ops framework for model and tracking evaluation, enabling reproducible benchmarking, regression detection, and data-driven model promotion.
- Migrated tracking from SORT to ByteTrack, improving accuracy by ~20% while preserving real-time, low-latency on-device inference.
- Created benchmarking and visualization tools that cut production failure modes by ~50% through systematic diagnostics.

Cisco Meraki | Software Engineering Intern, Edge AI Research

Paris, France | April 2022 – August 2022

- Built a real-time vehicle speed estimation pipeline in C++ and Python for Meraki MV cameras.

- Integrated Kalman-filter (SORT) tracking with YOLO detection for accurate on-device multi-object tracking.
- Optimized the inference pipeline for low-latency, low-power execution on embedded hardware.

Seamfix Limited | Software Engineer (QA & Automation)

Lagos, Nigeria | August 2020 – July 2021

- Built automation suites with Java, TestNG, and Appium, cutting regression cycle time and improving release quality.
- Performed API testing with REST-Assured and end-to-end regression across web, mobile, and desktop in an Agile environment.

Seamfix Limited | Software Engineering Intern

Lagos, Nigeria | May 2018 – October 2018

- Built automated tests for Android and web applications using Selenium and Appium.
- Contributed to enterprise Java backend development and explored Flutter for cross-platform mobile.

EDUCATION

École Polytechnique | Master of Science & Technology, Internet of Things: Innovation and Management

Palaiseau, France | 2021 – 2023

CGPA: 3.86 / 4.0 (Mention Bien)

University of Nigeria, Nsukka | B.Eng. in Electronic Engineering

Nsukka, Nigeria | 2014 – 2020

CGPA: 4.87 / 5.0 (First Class Honours)

HONORS & AWARDS

- Best Graduating Student, Faculty of Engineering, University of Nigeria, Nsukka (2018/2019).
- National Best Engineering Graduate (2nd Place), CODET Nigeria (2021).
- National Engineering Students' Project Competition (2nd Prize), 2019.

CERTIFICATIONS

- Advanced Embedded Linux Development Specialization, Coursera & University of Colorado Boulder.
- Machine Learning Specialization, Stanford University & DeepLearning.AI.
- Udacity ALX-T Full Stack Developer Nanodegree.